(f) Remove small screwdriver, or, factory alignment pin from housing and install plastic cover (two screws).

(7) Install CMP sensor to oil pump drive.

(8) Install sensor mounting bolt and tighten to 2 N·m (15 in. lbs.) torque.

(9) Connect CMP electrical connector to engine wiring harness.

(10) If removed, install spark plug and ignition coil rail.

(11) Connect DRB scan tool to data link connector. The data link connector is located in passenger compartment, below steering column.

(12) Gain access to "CAM/CRANK RELEARN" screen on DRB scan tool.

COIL RAIL - 4.0L

DESCRIPTION

A one-piece coil rail assembly containing three individual coils is used on the 4.0L 6–cylinder engine (Fig. 23). The coil rail must be replaced as one assembly. The bottom of the coil is equipped with 6 individual rubber boots (Fig. 23) to seal the 6 spark plugs to the coil. Inside each rubber boot is a spring. The spring is used for a mechanical contact between the coil and the top of the spark plug. These rubber boots and springs are a permanent part of the coil and are not serviced separately.

(1) The coil is bolted directly to the cylinder head (Fig. 24). One electrical connector (located at rear of coil) is used for all three coils.

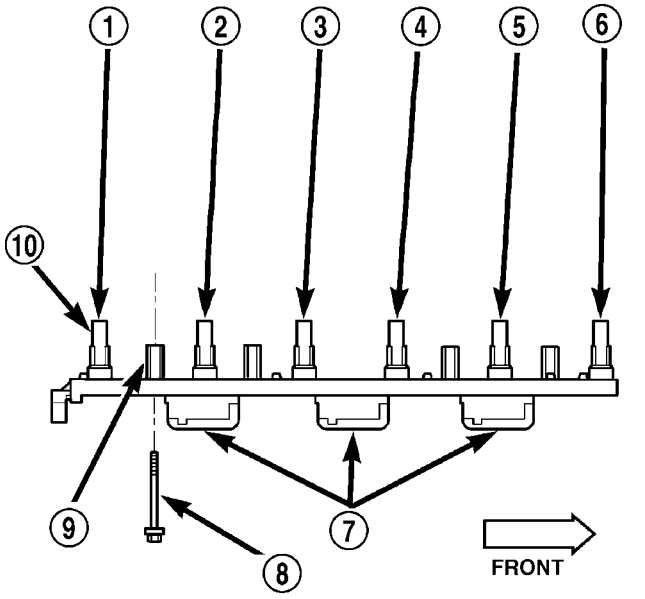

Fig. 23 Ignition Coil Assembly—4.0L 6–Cylinder Engine

1 - CYL. #6
2 - CYL. #5
3 - CYL. #4
4 - CYL. #3
5 - CYL. #2
6 - CYL. #1
7 - COILS (3)
8 - MOUNTING BOLTS (4)
9 - BOLT BASES (4)
10 - RUBBER BOOTS (6)

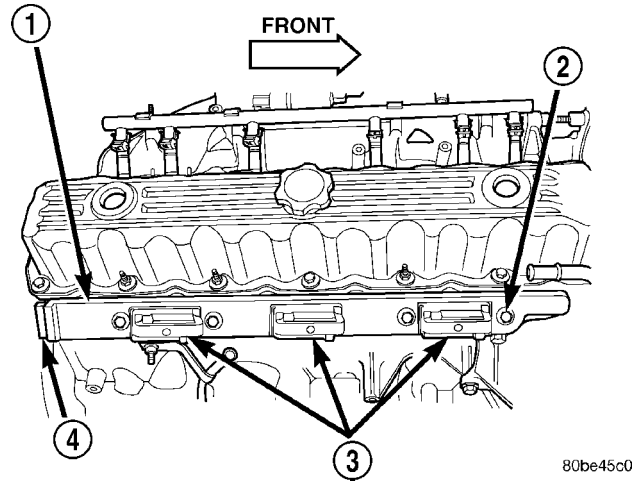

Fig. 24 Coil Location—4.0L Engine

1 - COIL RAIL
2 - COIL MOUNTING BOLTS (4)
3 - COIL
4 - COIL ELECTRICAL CONNECTION

OPERATION

Although cylinder firing order is the same as 4.0L Jeep engines of previous years, spark plug firing is not. The 3 coils dual-fire the spark plugs on cylinders 1-6, 2-5 and/or 3-4. When one cylinder is being fired